Endsem model solution EE322M

Q1(a) For example $x(t) = \alpha\,\text{sinc}^2(at) * \text{sinc}^3(bt)$, where α is a nonzero constant. Let $x_1(t) := \text{sinc}^2(at)$ and $x_2(t) := \text{sinc}^3(bt)$. Then $x(t) = x_1(t) * x_2(t) \Rightarrow X(j\omega) = X_1(j\omega)X_2(j\omega)$.

Note that

$$x_1(t) = \text{sinc}^2(at) = \left(\frac{\sin(a\pi t)}{a\pi}\right)^2 \Rightarrow x_1(t) = \frac{1-\cos(2a\pi t)}{2(a\pi)^2}$$
$$x_2(t) = \text{sinc}^3(bt) = \left(\frac{\sin(b\pi t)}{b\pi}\right)^3 \Rightarrow x_2(t) = \frac{3\sin(b\pi t) - \sin(3b\pi t)}{4(b\pi)^3}$$

Therefore, bandwidth of $x_1(t)$ is a Hz and that of $x_2(t)$ is $\frac{3b}{2}$ Hz. Since the spectrum of $x(t)$ is the product of $X_1(j\omega)$ and $X_2(j\omega)$, the bandwidth of $X(j\omega)$ is the minimum of a Hz and $\frac{3b}{2}$ Hz. Therefore, Nyquist sampling frequency is $2 \times \texttt{min}(a, \frac{3b}{2})$.

For example (Set-2): $a = 440$ and $b = 200$. Therefore, Nyquist sampling frequency is $2 \times \texttt{min}(440, \frac{3\times 200}{2}) = 2 \times \texttt{min}(440, 300) = 2 \times 300 = 600$ Hz.
For Set-1: Since both the terms are squared here, we will have Nyquist sampling frequency $2 \times \texttt{min}(440, 200) = 400$ Hz.

Set	f_s	**Set**	f_s	**Set**	f_s	**Set**	f_s	**Set**	f_s	**Set**	f_s
1	400 Hz	**2**	600 Hz	**3**	400 Hz	**4**	200 Hz	**5**	600 Hz	**6**	280 Hz
7	120 Hz	**8**	300 Hz	**9**	500 Hz	**10**	360 Hz	**11**	320 Hz	**12**	360 Hz
13	270 Hz	**14**	160 Hz	**15**	260 Hz	**16**	400 Hz	**17**	720 Hz	**18**	340 Hz
19	300 Hz	**20**	420 Hz	**21**	440 Hz	**22**	300 Hz	**23**	480 Hz		

Table 1: Solution of Q1(a)

Q1(b) Let the input be $x[n] = \sin(\omega_0 n + \phi)$ and the output be $y[n] = A\sin(\omega_0 n + \phi_1)$. Then we can rewrite the output as:

$$y[n] = A\sin(\omega_0(n - n_0) + \phi), \qquad \text{where } \omega_0 n_0 = \phi - \phi_1.$$

Therefore, the input and the output are related by the following relation:

$$y[n] = Ax[n - n_0] \Rightarrow Y(e^{j\omega}) = Ae^{-j\omega_0 n_0}X(e^{j\omega}) \Rightarrow \frac{Y(e^{j\omega})}{X(e^{j\omega})} = H(e^{j\omega}) = Ae^{-j\omega_0 n_0}.$$

Clearly, $\angle H(e^{j\omega})|_{\omega=\omega_0} = -\omega_0 n_0$.

However, there is no information about the other frequencies. Hence, a generalized formula for $\angle H(e^{j\omega})$ that applies to all frequencies cannot be formulated.

Set	$\angle H(e^{j\omega_0})$	**Set**	$\angle H(e^{j\omega_0})$	**Set**	$\angle H(e^{j\omega_0})$	**Set**	$\angle H(e^{j\omega_0})$	**Set**	$\angle H(e^{j\omega_0})$	**Set**	$\angle H(e^{j\omega_0})$
1	-300	**2**	-400	**3**	-30	**4**	-230	**5**	-300	**6**	-300
7	-300	**8**	-100	**9**	-120	**10**	-400	**11**	-300	**12**	-300
13	-300	**14**	-200	**15**	-300	**16**	-300	**17**	-300	**18**	-140
19	-300	**20**	-40	**21**	-300	**22**	-300	**23**	-300		

Table 2: Solution of Q1(b) only for frequency ω_0

Q1(c) Let $x[n] = \left(\frac{1}{a}\right)^n u[n]$. Therefore, $y[n] = x^2[n] = \left(\frac{1}{a}\right)^{2n} u[n] = \left(\frac{1}{a^2}\right)^n u[n]$. Using the formula for DTFT, we have

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{a^2}\right)^n u[n] e^{-j\omega n} = \sum_{n=0}^{\infty} \left(\frac{1}{a^2}\right)^n e^{-j\omega n}. \tag{1}$$

Evaluating equation (1) at $\omega = 0$, we have

$$Y(e^{j0}) = \sum_{n=0}^{\infty} \left(\frac{1}{a^2}\right)^n \Rightarrow Y(e^{j0}) = \frac{1}{1-\frac{1}{a^2}} = \frac{a^2}{a^2-1}.$$

For example (Set-1): $a = 4$ and therefore $Y(e^{j0}) = \frac{16}{15}$.

Set	$Y(e^{j0})$	**Set**	$Y(e^{j0})$	**Set**	$Y(e^{j0})$	**Set**	$Y(e^{j0})$	**Set**	$Y(e^{j0})$	**Set**	$Y(e^{j0})$
1	$\frac{16}{15}$	**2**	$\frac{81}{80}$	**3**	$\frac{36}{35}$	**4**	$\frac{49}{48}$	**5**	$\frac{49}{48}$	**6**	$\frac{36}{35}$
7	$\frac{49}{48}$	**8**	$\frac{64}{63}$	**9**	$\frac{81}{80}$	**10**	$\frac{100}{99}$	**11**	$\frac{121}{120}$	**12**	$\frac{144}{143}$
13	$\frac{169}{168}$	**14**	$\frac{196}{195}$	**15**	$\frac{225}{224}$	**16**	$\frac{256}{255}$	**17**	$\frac{289}{288}$	**18**	$\frac{324}{323}$
19	$\frac{361}{360}$	**20**	$\frac{400}{399}$	**21**	$\frac{441}{440}$	**22**	$\frac{484}{483}$	**23**	$\frac{529}{528}$		

Table 3: Solution of Q1(c)

Q1(d) $x(t) = \sin(2\pi f t) + \cos(2\pi f t) = \sqrt{2}\sin\left(2\pi f t + \frac{\pi}{4}\right)$.
Therefore, Nyquist sampling frequency is $f_s = 2 \times f$. The peak-to-peak amplitude of $x(t)$ is $2\sqrt{2}$. If step-size of quantization is Δ, then the number of levels $L = \frac{2\sqrt{2}}{\Delta}$. Therefore the number of bits N must satisfy $2^N = L \Rightarrow N = \log_2 L$. Therefore, data rate in bits per second is $f_s \times N$.
For example (Set-1): $f = 100 \Rightarrow f_s = 200$ and $\Delta = 0.75$. Therefore, $L = \frac{2\sqrt{2}}{0.75} = 3.77$. Number of levels have to be positive integers. We cannot take 3 because this might leave out some information from the signal. Hence, we always choose the next largest non-negative integer that can be written as 2^x, where x is a non-negative integer. We go for 2^x because the number of levels are given by 2^N. In this case we take $L = 4$ and therefore $N = \log_2 L = 2$. Therefore, the data rate is $200 \times 2 = 400$ bits/sec.

1(e) Let $x(t) = \frac{1}{a+j2\pi t}$. Note that the Fourier transform of $e^{-at}u(t)$ is $\frac{1}{a+j\omega} = \frac{1}{a+j2\pi f}$. Hence, we can use duality here.

From the duality property of Fourier transform, we have

$$\mathcal{F}(g(t)) = G(f) \Leftrightarrow \mathcal{F}(G(t)) = g(-f).$$

Therefore, $\mathcal{F}\left(\frac{1}{a+j2\pi t}\right) = e^{af}u(-f)$.
For example (Set-1): $a = 4$. Therefore, $\mathcal{F}\left(\frac{1}{4+j2\pi t}\right) = e^{4f}u(-f)$.

Set	f_s	Δ	L	N	Data rate (bits/sec)	**Set**	f_s	Δ	L	N	Data rate (bits/sec)	**Set**	f_s	Δ	L	N	Data rate (bits/sec)
1	200	0.75	4	2	400	**2**	200	0.50	8	3	600	**3**	100	0.75	4	2	200
4	75	1	4	2	150	**5**	70	0.75	4	2	140	**6**	110	0.50	8	3	330
7	180	0.75	4	2	360	**8**	120	0.25	16	4	480	**9**	250	1	4	2	500
10	120	0.50	8	3	360	**11**	160	0.75	4	2	320	**12**	180	0.50	8	3	540
13	90	1	4	2	180	**14**	80	0.75	4	2	160	**15**	130	0.50	8	3	390
16	140	0.75	4	2	280	**17**	320	1	4	2	640	**18**	170	0.75	4	2	340
19	180	0.50	8	3	540	**20**	210	0.50	8	3	630	**21**	220	0.50	8	3	660
22	220	1	4	2	440	**23**	240	1	4	2	480						

Table 4: Solution of Q1(d)

Set	**CTFT**	**Set**	**CTFT**	**Set**	**CTFT**	**Set**	**CTFT**	**Set**	**CTFT**
1	$e^{4f}u(-f)$	**2**	$e^{9f}u(-f)$	**3**	$e^{3f}u(-f)$	**4**	$e^{3f}u(-f)$	**5**	$e^{7f}u(-f)$
6	$e^{6f}u(-f)$	**7**	$e^{7f}u(-f)$	**8**	$e^{8f}u(-f)$	**9**	$e^{9f}u(-f)$	**10**	$e^{10f}u(-f)$
11	$e^{11f}u(-f)$	**12**	$e^{12f}u(-f)$	**13**	$e^{13f}u(-f)$	**14**	$e^{14f}u(-f)$	**15**	$e^{15f}u(-f)$
16	$e^{16f}u(-f)$	**17**	$e^{17f}u(-f)$	**18**	$e^{18f}u(-f)$	**19**	$e^{4f}u(-f)$	**20**	$e^{20f}u(-f)$
21	$e^{21f}u(-f)$	**22**	$e^{22f}u(-f)$	**23**	$e^{23f}u(-f)$				

Table 5: Solution of Q1(e)

Q(2) $G(s) = \frac{1}{s+a} + \frac{1}{s+b}$. Therefore, $g(t) = \left(e^{-at} + e^{-bt}\right) u(t)$.
Sampling the impulse response with a sampling period of T_s gives:

$$g(nT_s) = \left[e^{-anT_s} + e^{-bnT_s}\right] u(nT_s)$$

Therefore,

$$G(e^{j\omega}) = \sum_{n=-\infty}^{\infty} g(nT_s)e^{-j\omega n} = \sum_{n=0}^{\infty} \left[e^{-anT_s} + e^{-bnT_s}\right] e^{-j\omega n}$$

Since we are interested in $\omega = 0$, we find $G(e^{j0})$ from the above expression:

$$G(e^{j0}) = \sum_{n=0}^{\infty} \left[e^{-anT_s} + e^{-bnT_s}\right] = \frac{1}{1-e^{-aT_s}} + \frac{1}{1-e^{-bT_s}}$$

Let $e^{-T_s} = x$. Further, let the desired gain at 0 rad/sec be β dB. Then, we have

$$20\log|G(e^{j0})| = \beta \Rightarrow |G(e^{j0})| = 10^{\beta/20} \Rightarrow \frac{1}{1-x^a} + \frac{1}{1-x^b} = 10^{\beta/20}. \tag{2}$$

For example (Set-1): $a = 1$, $b = 2$, and $\beta = 9$. Therefore,

$$\frac{1}{1-x} + \frac{1}{1-x^2} = 10^{\frac{9}{20}} \Rightarrow x = -0.745,\ 0.39.$$

x cannot be negative, since e^{-T_s} is never negative. Hence, $x = 0.39 \Rightarrow e^{-T_s} = 0.39 \Rightarrow -T_s = -0.942 \Rightarrow T_s = 0.942 \Rightarrow f_s = 1.06$ Hz.

Note that x cannot be complex in equation (2), since this won't give us real-valued T_s. Hence, we look only for real roots of equation (2).

Set	Real roots*		T_s (in sec)	f_s (in Hz)	Set	Real roots*		T_s (in sec)	f_s (in Hz)	Set	Real roots*		T_s (in sec)	f_s (in Hz)
1	-0.745	0.39	0.942	1.06	**9**	0.537		0.622	1.61	**17**	No real roots		Not possible	
2	-0.723	0.596	0.518	1.93	**10**	0.537	-0.954	0.622	1.61	**18**	-0.642	-0.18	Not possible	
3	0.424		0.858	1.17	**11**	0.685	-0.685	0.378	2.64	**19**	-0.167		Not possible	
4	-0.496	-0.005	Not possible		**12**	0.7	-0.76	0.357	2.80	**20**	-0.792	-0.168	Not possible	
5	No real roots		Not possible		**13**	0.71	-0.71	0.342	2.92	**21**	No real roots		Not possible	
6	No real roots		Not possible		**14**	-0.749	0.716	0.334	2.99	**22**	No real roots		Not possible	
7	-0.842	0.591	0.526	1.90	**15**	No real roots		Not possible		**23**	No real roots		Not possible	
8	-0.865	-0.005	Not possible		**16**	0.724	-0.742	0.323	3.10					

Table 6: Solution of Q(2) – the ones in red are not acceptable as e^{-T_s} cannot be negative.

Q(3) The CTFT of the output $y(t)$ is given by

$$Y(j\omega) = \int_{-\infty}^{\infty} y(t)e^{-j\omega t}dt.$$

Therefore,

$$Y(j0) = \int_{-\infty}^{\infty} = \text{Area under the } y(t) \text{ plot} = \alpha \text{ (a non-zero real number)}.$$

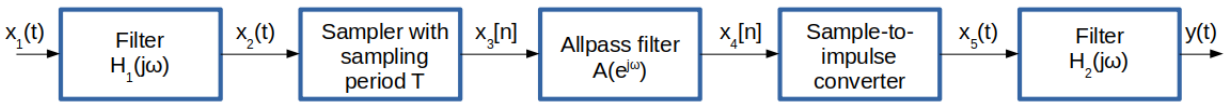
From the specification of the bandpass filter it is clear that $H(j0) = 0$.

For an LTI system, the output is given by

$$y(t) = x(t) * h(t) \Rightarrow Y(j\omega) = X(j\omega)H(j\omega) \Rightarrow Y(j0) = X(j0)H(j0) \Rightarrow \alpha = X(j0) \times 0.$$

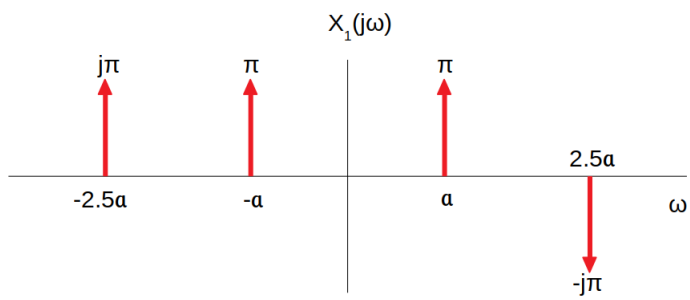
This is not true for any $x(t)$. Hence, it is not possible to recover the input.

Q4(a) The block diagram of the system given is as follows:



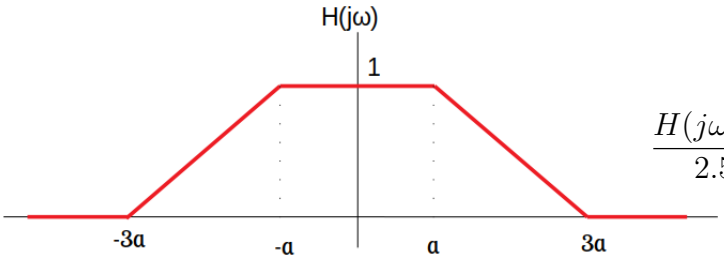
Q4(b) Consider that the input is $x_1(t) = \cos(at) + \sin(2.5at)$. The CTFT of $x_(t)$ is

$$X_1(j\omega) = \pi\left[\delta(\omega + a) + \delta(\omega - a)\right] + \frac{\pi}{j}\left[\delta(\omega + 2.5a) - \delta(\omega - 2.5a)\right]$$

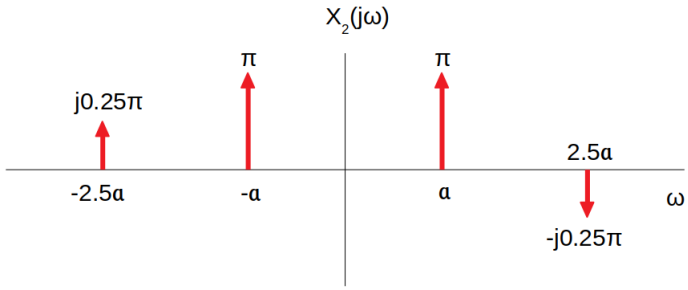


Since $x_2(t) = x_1(t) * h_1(t)$, $X_2(j\omega) = X_1(j\omega)H_1(j\omega)$. We will need to find the gain of the filter at $\omega = 2.5a$.

The gain of the filter at $\omega = 2.5a$ is 0.25 and at $\omega = a$ it is 1. Therefore, $X_2(j\omega)$ becomes:



$$\frac{H(j\omega)|_{2.5a} - 1}{2.5a - a} = \frac{0-1}{3a-a} \Rightarrow H(j\omega)|_{2.5a} = 0.25.$$



Evidently,

$$X_2(j\omega) = \pi\left[\delta(\omega + a) + \delta(\omega - a)\right] + 0.25\frac{\pi}{j}\left[\delta(\omega + 2.5a) - \delta(\omega - 2.5a)\right]$$
$$\Rightarrow x_2(t) = \cos(at) + 0.25\sin(2.5at).$$

On sampling $x_2(t)$ with a sampling period of T, we have

$$x_2(nT) = \cos(anT) + 0.25\sin(2.5anT) \Rightarrow x_3[n] = \cos[an] + 0.25\sin[2.5an].$$

Next we have an allpass filter and it allows all frequency to pass through with unity gain.

$$x_4[n] = x_3[n] * a[n] \Rightarrow X_4(e^{j\omega}) = X_3(e^{j\omega})A(e^{j\omega}) = X_3(e^{j\omega}) \Rightarrow x_4[n] = x_3[n].$$

Therefore, $x_4[n] = \cos[an] + 0.25\sin[2.5an]$. Now,

$$x_5(t) = \sum_{n=-\infty}^{\infty} x_4[n]\delta(t - nT) \Rightarrow x_5(t) = \sum_{n=-\infty}^{\infty} \left(\cos[an] + 0.25\sin[2.5an]\right)\delta(t - nT).$$

Next we look at the spectrum of $x_5(t)$. Note that

$$\begin{aligned}
x_5(t) &= \sum_{n=-\infty}^{\infty} \left(\cos[an] + 0.25\sin[2.5an]\right)\delta(t - nT) \\
&= \sum_{n=-\infty}^{\infty} \left(\cos(anT) + 0.25\sin(2.5anT)\right)\delta(t - nT) \\
&= \underbrace{\left(\cos(at) + 0.25\sin(2.5at)\right)}_{x_2(t)} \sum_{n=-\infty}^{\infty} \delta(t - nT) \\
&= x_2(t)p(t), \quad \text{where } p(t) := \sum_{n=-\infty}^{\infty} \delta(t - nT).
\end{aligned}$$

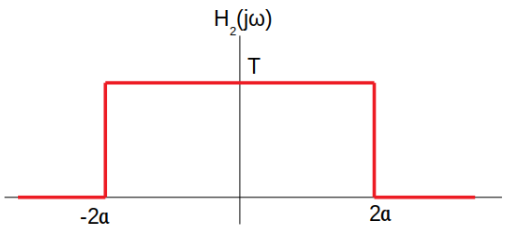
Therefore,

$$X_5(j\omega) = \frac{1}{2\pi}\left[X_2(j\omega) * P(j\omega)\right] = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_2(j(\omega - k\omega_s)), \qquad \text{where } \omega_s = \frac{2\pi}{T} \tag{3}$$

Note that $a = \frac{\pi}{2T}$ in your question. Therefore, $4a = \frac{2\pi}{T} = \omega_s$. Using this in equation (3) we have

$$\begin{aligned}
X_5(j\omega) &= \frac{1}{T}\sum_{k=-\infty}^{\infty} X_2\left(j(\omega - 4ak)\right) \\
&= \frac{1}{T}\sum_{k=-\infty}^{\infty} \Big[\pi\Big(\delta(\omega + a - 4ak) + \delta(\omega - a - 4ak)\Big) \\
&\qquad + 0.25\frac{\pi}{j}\Big(\delta(\omega + 2.5a - 4ak) - \delta(\omega - 2.5a - 4ak)\Big)\Big].
\end{aligned}$$

Note that $y(t) = x_5(t) * h_2(t) \Rightarrow Y(j\omega) = X_5(j\omega)H_2(j\omega)$ and the $H_2(j\omega)$ is as follows:



Any frequency component above $2a$ will be fully attenuated when passed through the last filter $H_2(j\omega)$. Hence, the CTFT of the output is

$$\begin{aligned}
Y(j\omega) &= T \times \frac{1}{T}\left[\pi\Big(\delta(\omega + a) + \delta(\omega - a)\Big) + 0.25\frac{\pi}{j}\Big(\delta(\omega - 1.5a) - \delta(\omega + 1.5a)\Big)\right] \\
&= \pi\Big(\delta(\omega + a) + \delta(\omega - a)\Big) - 0.25\frac{\pi}{j}\Big(\delta(\omega + 1.5a) - \delta(\omega - 1.5a)\Big)
\end{aligned}$$

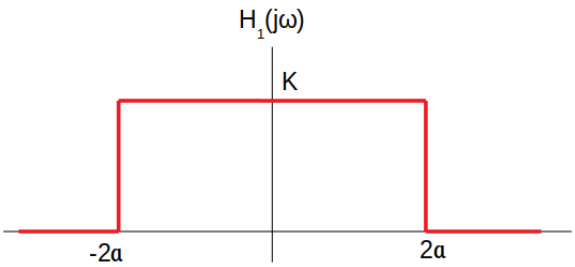
Therefore,

$$y(t) = \cos(at) - \frac{1}{4}\sin(1.5at).$$

Set	a	**Set**	a	**Set**	a	**Set**	a	**Set**	a
1	100	**2**	50	**3**	25	**4**	200	**5**	70
6	110	**7**	60	**8**	150	**9**	250	**10**	120
11	160	**12**	180	**13**	90	**14**	80	**15**	130
16	140	**17**	160	**18**	170	**19**	180	**20**	210
21	220	**22**	230	**23**	240				
$x_2(t) = \cos(at) + 0.25\sin(2.5at)$									
$x_3[n] = \cos[an] + 0.25\sin[2.5an]$									
$x_4[n] = \cos[an] + 0.25\sin[2.5an]$									
$x_5(t) = \sum_{n=-\infty}^{\infty}\left(\cos[an] + 0.25\sin[2.5an]\right)\delta(t - n\frac{\pi}{2a})$									
$y(t) = \cos(at) - 0.25\sin(1.5at)$									

Table 7: Solution of Q(4)

Q4(c) The frequency response of the low pass filter is $H_1(j\omega)$ is –



We have assumed K to be the gain of the filter $H_1(j\omega)$ (the gain of the filter is not mentioned in the question). Evidently, the output of the filter $H_1(j\omega)$ will not have frequency components greater than $2a$. Hence,

$$X_2(j\omega) = K\pi\left(\delta(\omega + a) + \delta(\omega - a)\right) \Rightarrow x_2(t) = K\cos(at)$$

Therefore, from Q4(b) if we remove the contribution due to the sine term we have our output.

$$Y(j\omega) = K\pi\left(\delta(\omega + a) + \delta(\omega - a)\right) \Rightarrow y(t) = K\cos(at).$$

Q4(d) Since the digital filter $A(e^{j\omega})$ has a cutoff frequency at π, it allows all signals from $-\pi$ to π unattenuated. Since it is a digital filter, the frequency response is periodic with a period of 2π. Therefore, the new filter is also an allpass filter. We have already found the output in Q4(b).

Q5(a) Can be computed easily (answer varies for all sets). Modify the code that you wrote for Question 10 in Assignment-4 to check your answers. You can also use `fft` command in SCILAB or MATLAB to check the answer – ofcourse in that case you might not get the output of the first and second stage of the butterfly.

Q5(b) You can use either the DIT radix-2 FFT algorithm or the DIF radix-2 FFT algorithm: see Figures 1 and 2.

Q5(c) The number of complex multiplication $= \frac{N}{2}\log_2 N = \frac{8}{2}\log_2 8 = 12$.
The number of complex addition $= N\log_2 N = 8\log_2 8 = 24$.
Let time taken for complex multiplication and complex addition be α sec and β sec, respectively. Therefore, total time taken for FFT computation is $12\alpha + 24\beta$.

Set	**Time taken** (μsec)	**Set**	**Time taken** (μsec)	**Set**	**Time taken** (μsec)	**Set**	**Time taken** (μsec)	**Set**	**Time taken** (μsec)
1	48	**2**	36	**3**	60	**4**	48	**5**	42
6	48	**7**	48	**8**	72	**9**	48	**10**	144
11	48	**12**	120	**13**	168	**14**	72	**15**	144
16	96	**17**	72	**18**	48	**19**	120	**20**	144
21	48	**22**	36	**23**	96				

Table 8: Solution of Q5(c)

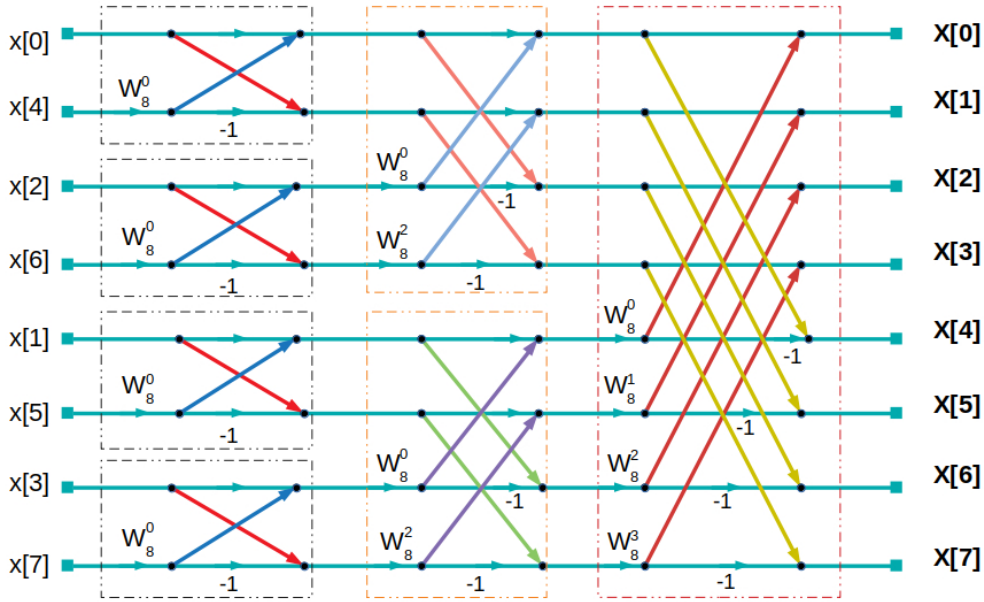


Figure 1: Q5(b): Radix-2 DIT FFT algorithm

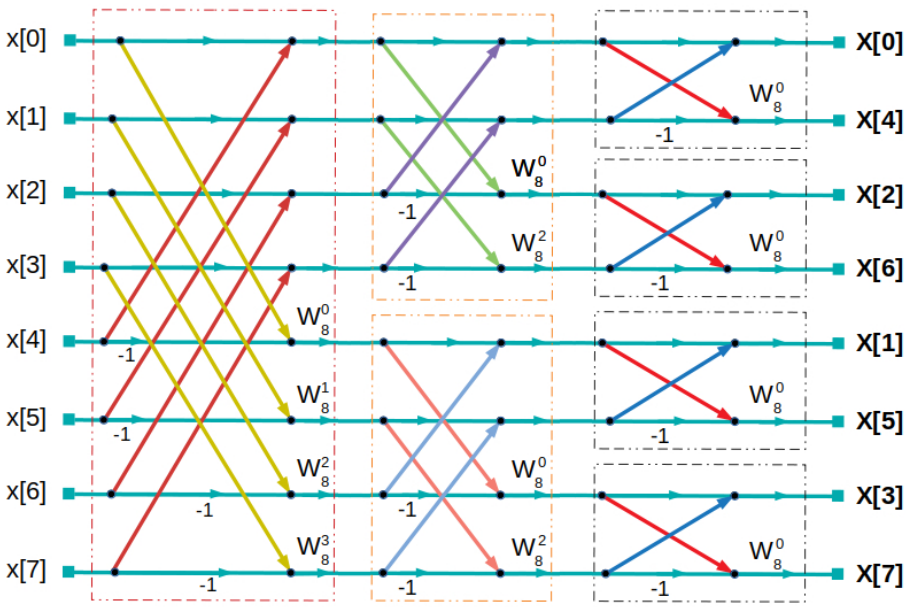


Figure 2: Q5(b): Radix-2 DIF FFT algorithm

Q5(d) A trivial solution is $x[n] = \{0, 0, 0, 0, 0, 0, 0, 0\}$.
Side note: You can also find nontrivial solutions. Let $\mathbb{W}_8$ be the matrix corresponding to the 8-point DFT as given below:

$$\mathbb{W} := \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & W_8 & W_8^2 & W_8^3 & W_8^4 & W_8^5 & W_8^6 & W_8^7 \\ 1 & W_8^2 & W_8^4 & W_8^6 & W_8^8 & W_8^{10} & W_8^{12} & W_8^{14} \\ 1 & W_8^3 & W_8^6 & W_8^9 & W_8^{12} & W_8^{15} & W_8^{18} & W_8^{21} \\ 1 & W_8^4 & W_8^8 & W_8^{12} & W_8^{16} & W_8^{20} & W_8^{24} & W_8^{28} \\ 1 & W_8^5 & W_8^{10} & W_8^{15} & W_8^{20} & W_8^{25} & W_8^{30} & W_8^{35} \\ 1 & W_8^6 & W_8^{12} & W_8^{18} & W_8^{24} & W_8^{30} & W_8^{36} & W_8^{42} \\ 1 & W_8^7 & W_8^{14} & W_8^{21} & W_8^{28} & W_8^{35} & W_8^{42} & W_8^{49} \end{bmatrix}$$

We want a signal x such that $\mathbb{W}x = x \Rightarrow (\mathbb{I} - \mathbb{W})x = 0$. It is always possible to find a vector x that satisfies the equation (since $\det(\mathbb{I}-\mathbb{W})$ is zero); infact you can find infinitely many vectors that satisfies the equation. You can use the command `kernel`$(\mathbb{I} - \mathbb{W})$ in SCILAB or `null`$(\mathbb{I} - \mathbb{W})$ in MATLAB to find one such vector.

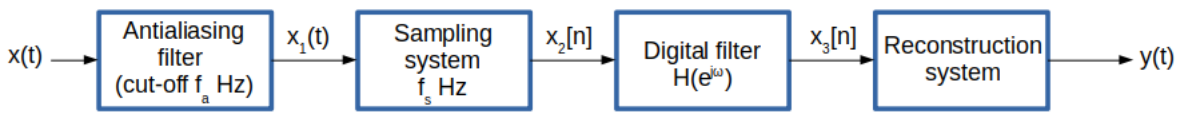
Q5(e) $h[n] = \{a, b, c\}$ and $x[n] = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8\}$. If we choose $N = 4$ and $M = 3$, length of the blocks L is given by $N = L + M - 1 \Rightarrow L = N - M + 1 = 2$.

Everyone will have different steps depending on what you choose for N. Finally the output would be:

Set	Output	Set	Output	Set	Output
1	$\{5, 11, 9, 7, 9, 11, 15, 21, 18, 6\}$	**9**	$\{1, 3, 5, 7, 9, 11, 13, 15, 12, 4\}$	**17**	$\{1, 3, 4, 4, 6, 10, 12, 12, 9, 3\}$
2	$\{6, 11, 12, 8, 5, 7, 8, 12, 9, 6\}$	**10**	$\{2, 4, 5, 4, 5, 7, 10, 11, 8, 4\}$	**18**	$\{2, 4, 5, 4, 5, 7, 7, 5, 2, 1\}$
3	$\{2, 6, 10, 11, 8, 7, 9, 11, 9, 3\}$	**11**	$\{4, 8, 10, 8, 5, 4, 8, 13, 12, 6\}$	**19**	$\{1, 4, 6, 6, 7, 7, 6, 6, 4, 1\}$
4	$\{3, 12, 21, 23, 17, 9, 7, 9, 8, 3\}$	**12**	$\{-1, -2, 4, 5, -3, 3, 8, 6, 4, 1\}$	**20**	$\{4, 4, 7, 5, 5, 6, 5, 7, 3, 2\}$
5	$\{1, 3, 6, 9, 12, 15, 18, 21, 15, 8\}$	**13**	$\{-1, -4, -5, -5, 0, 15, 20, 20, 16, 4\}$	**21**	$\{2, 6, 11, 16, 13, 10, 11, 16, 11, 4\}$
6	$\{8, 17, 12, 8, 12, 13, 14, 16, 10, 2\}$	**14**	$\{-1, 2, 8, 10, 10, 10, 8, 6, 7, 4\}$	**22**	$\{3, 10, 16, 13, 7, 6, 9, 14, 14, 8\}$
7	$\{2, 10, 11, 11, 6, 9, 8, 11, 7, 6\}$	**15**	$\{2, 3, 6, 5, 6, 4, 6, 5, 4, 1\}$	**23**	$\{2, 7, 12, 13, 11, 10, 7, 8, 6, 4\}$
8	$\{6, 19, 28, 24, 12, 8, 12, 16, 14, 5\}$	**16**	$\{-1, 0, 5, 8, 8, 8, 8, 5, 0, -1\}$		

Table 9: Solution of Q5(e)

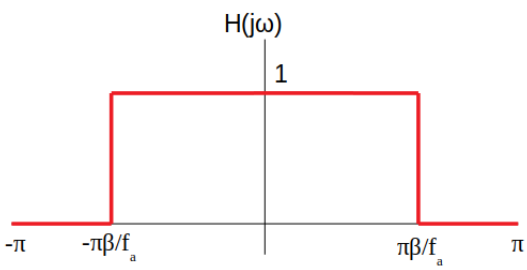
Q6(a) The system is given by:



1. Anti-aliasing filter: We choose an anti-aliasing filter with cut-off frequency at f_a Hz.
2. Sampling system: The sampling frequency of the sampler should be greater than $f_s = 2f_a$. We choose the Nyquist sampling rate for the sampler.
3. Digital filter:

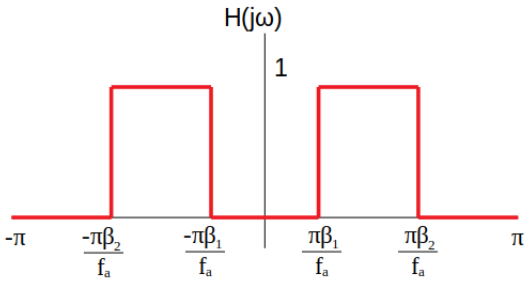
– Suppose that we want the output to not have frequency above γ Hz, i.e., we want $f \leqslant \gamma$. In order to achieve this with a digital filter, we need to find out to what digital frequency does the continuous frequency γ correspond to. The relation between continuous frequency f and digital frequency Ω is: $\Omega = 2\pi\frac{f}{f_s}$. Therefore,

$$f \leqslant \gamma \Rightarrow \frac{\Omega f_s}{2\pi} \leqslant \gamma$$
$$\Rightarrow \Omega \leqslant \frac{2\pi\gamma}{f_s} \Rightarrow \Omega \leqslant \frac{\pi\gamma}{f_a}$$



– Suppose that we want the output to have frequency between γ_1 Hz and γ_2 Hz, i.e., we want $\gamma_1 \leqslant f \leqslant \gamma_2$. Then

$$\gamma_1 \leqslant f \leqslant \gamma_2 \Rightarrow \gamma_1 \leqslant \frac{\Omega f_s}{2\pi} \leqslant \gamma_2 \Rightarrow \frac{\pi\gamma_1}{f_a} \leqslant \Omega \leqslant \frac{\pi\gamma_2}{f_a}.$$



Q6(b) Depends on the question.
Q(6) is a system design problem and answer would vary based on your choice of blocks. Hence, no specific answer can be given for each set.

**********THE END**********